

4 IMPORTANT CONCEPTS

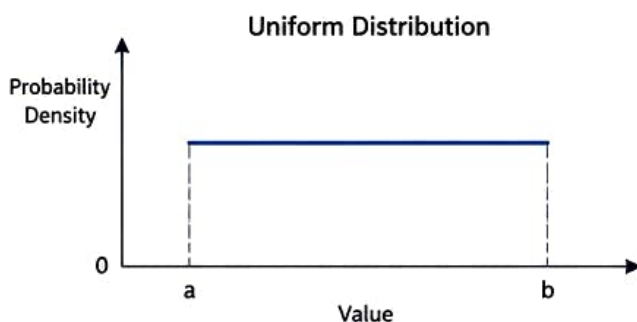
Statistics & AI / ML Basics

1. UNIFORM DISTRIBUTION

- In Uniform Distribution, every value in a given range has an equal chance of occurring.
- The graph looks like a rectangle.

Example:

Rolling a fair dice – 1, 2, 3, 4, 5, 6 all have the same probability (1/6).



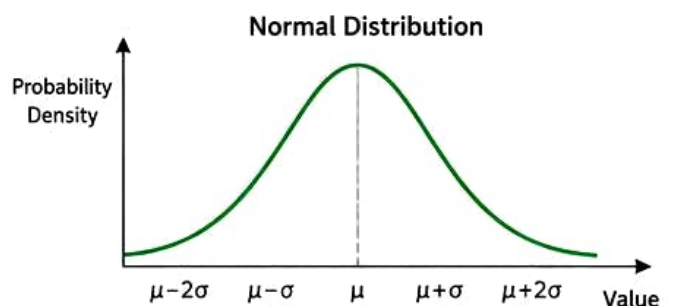
Key Idea: Equal chance for all values in the range $[a, b]$.

2. NORMAL DISTRIBUTION

- In Normal Distribution, values near the mean (average) occur most frequently.
- The graph is bell-shaped.
- It is symmetric around the mean.

Example:

Heights of people, exam marks, measurement errors, etc.



Key Idea: Most values are around the mean (μ), fewer as we move away.

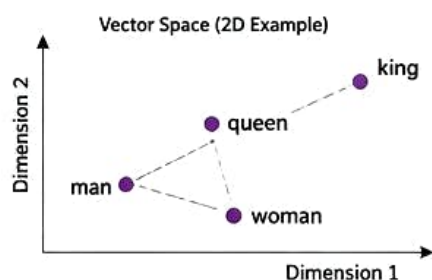
3. WORD TO VECTOR

- Words are converted into numbers (vectors) so that machines can understand them.
- Similar words have similar vectors.

Example:

Using Word Embedding (like Word2Vec)

Words	Vectors (Example)
king	[0.21, -0.35, 0.12, ...]
queen	[0.20, -0.31, 0.11, ...]
man	[0.19, -0.20, 0.10, ...]
woman	[0.18, -0.18, 0.09, ...]



Key Idea: Words $\rightarrow$ Numbers (vectors) so that meaning and similarity can be captured.

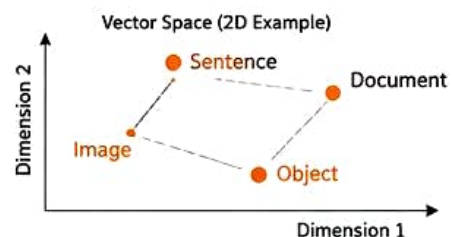
4. SPACE TO VECTOR

- Items like documents, sentences, images, or objects are converted into vectors.
- These vectors capture the information in a mathematical form.

Example:

Representing a sentence in vector form using embeddings.

Space (Examples)	Vectors (Example)
Document	[0.12, -0.08, 0.34, ...]
Sentence	[0.15, -0.11, 0.28, ...]
Image	[0.21, -0.25, 0.31, ...]
Object	[0.17, -0.19, 0.27, ...]



Key Idea: Any data from the real world (space) can be represented as vectors.



Why is this important?

These concepts are the foundation of Data Science, Machine Learning and Generative AI. They help machines understand data, find patterns and make intelligent decisions.

